

1.CREATE TABLE

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow_db

- books
- donation_history
- loans
- members
- Views
- Stored Procedures
- Functions

sys

Administration Schemas

Information

No object selected

library week1 week2 DDL*

Limit to 1000 rows

```
1 USE bookflow_db;
2
3 CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
```

Output

Action Output

#	Time	Action	Message
1	15:12:09	USE bookflow_db	0 row(s) affected
2	15:12:11	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected

2.ALTER TABLE — ADD COLUMN

MySQL Workbench

Local instance MySQL80

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Navigator

SCHEMAS

Filter objects

bookflow_db

- books
- donation_history
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- Views
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- Functions

sys

Administration Schemas

Information

No object selected

library week1 week2 DDL*

Limit to 1000 rows

```
8 amount DECIMAL(10,2),
9 transaction_date DATE
10 );
11
12 ALTER TABLE bank_transactions
13 ADD account_no VARCHAR(20);
14 DESCRIBE bank_transactions;
```

Result Grid

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI		
customer_name	varchar(50)	YES			
branch_name	varchar(50)	YES			
transaction_type	varchar(20)	YES			
amount	decimal(10,2)	YES			
transaction_date	date	YES			
account_no	varchar(20)	YES			

Result 1 x

Output

Action Output

#	Time	Action	Message
1	15:12:09	USE bookflow_db	0 row(s) affected
2	15:12:11	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
3	15:21:27	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected
4	15:21:39	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	Error Code: 1060, D
5	15:21:42	DESCRIBE bank_transactions	7 row(s) returned

3.ALTER TABLE — MODIFY COLUMN

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator library week1 week2 DDL* x Limit to 1000 rows

SCHEMAS

Filter objects

bookflow_db

- books
- donation_history
- loans
- members
- Views
- Stored Procedures
- Functions

sys

Administration Schemas

Information

No object selected

```
12 ALTER TABLE bank_transactions
13 ADD account_no VARCHAR(20);
14 DESCRIBE bank_transactions;
15
16 ALTER TABLE bank_transactions
17 MODIFY customer_name VARCHAR(100);
18 DESCRIBE bank_transactions;
```

Result Grid

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI		
customer_name	varchar(100)	YES			
branch_name	varchar(50)	YES			
transaction_type	varchar(20)	YES			
amount	decimal(10,2)	YES			
transaction_date	date	YES			
account_no	varchar(20)	YES			

Result 2 x

Output

Action Output

#	Time	Action	Message
2	15:12:11	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected
3	15:21:27	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Dup
4	15:21:39	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	Error Code: 1060. Duplicate column
5	15:21:42	DESCRIBE bank_transactions	7 row(s) returned
6	15:25:48	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Dup
7	15:26:05	DESCRIBE bank_transactions	7 row(s) returned

4. RENAME TABLE

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator library week1 week2 DDL* x Limit to 1000 rows

SCHEMAS

Filter objects

bookflow_db

- books
- donation_history
- loans
- members
- Views
- Stored Procedures
- Functions

sys

Administration Schemas

Information

No object selected

```
16 ALTER TABLE bank_transactions
17 MODIFY customer_name VARCHAR(100);
18 DESCRIBE bank_transactions;
19
20 RENAME TABLE bank_transactions
21 TO customer_transactions;
22 SHOW TABLES;
```

Result Grid

Tables_in_bookflow_db
books
customer_transactions
donation_history
loans
members

Result 3 x

Output

Action Output

#	Time	Action	Message
4	15:21:39	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	
5	15:21:42	DESCRIBE bank_transactions	
6	15:25:48	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	
7	15:26:05	DESCRIBE bank_transactions	
8	15:27:08	RENAME TABLE bank_transactions TO customer_transactions	
9	15:27:17	SHOW TABLES	

5. TRUNCATE TABLE

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'bookflow_db' expanded, showing tables like 'books', 'donation_history', 'loans', and 'members'. The main editor window contains the following SQL script:

```
19
20 • RENAME TABLE bank_transactions
21   TO customer_transactions;
22 • SHOW TABLES;
23
24 • TRUNCATE TABLE customer_transactions;
25 • DESCRIBE customer_transactions;
```

The 'Result Grid' shows the output of the SQL execution:

Field	Type	Null	Key	Default	Extra
bn_id	int	NO	PRI		UNSIGNED
customer_name	varchar(100)	YES			UNSIGNED
branch_name	varchar(50)	YES			UNSIGNED
transaction_type	varchar(20)	YES			UNSIGNED
amount	decimal(10,2)	YES			UNSIGNED
transaction_date	date	YES			UNSIGNED
account_no	varchar(20)	YES			UNSIGNED

The 'Output' pane shows the 'Action Output' table:

#	Time	Action	Message
6	15:25:48	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected
7	15:26:05	DESCRIBE bank_transactions	7 row(s) returned
8	15:27:08	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
9	15:27:17	SHOW TABLES	5 row(s) returned
10	15:31:30	TRUNCATE TABLE customer_transactions	0 row(s) affected
11	15:31:36	DESCRIBE customer_transactions	7 row(s) returned

6. DROP Table

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'bookflow_db' expanded, showing tables like 'books', 'donation_history', 'loans', and 'members'. The main editor window contains the following SQL script:

```
22 • SHOW TABLES;
23
24 • TRUNCATE TABLE customer_transactions;
25 • DESCRIBE customer_transactions;
26
27 • DROP TABLE customer_transactions;
28 • SHOW TABLES;
```

The 'Result Grid' shows the output of the SQL execution:

Field	Type	Null	Key	Default	Extra
bn_id	int	NO	PRI		UNSIGNED
customer_name	varchar(100)	YES			UNSIGNED
branch_name	varchar(50)	YES			UNSIGNED
transaction_type	varchar(20)	YES			UNSIGNED
amount	decimal(10,2)	YES			UNSIGNED
transaction_date	date	YES			UNSIGNED
account_no	varchar(20)	YES			UNSIGNED

The 'Output' pane shows the 'Action Output' table:

#	Time	Action	Message
1	12:12:20	USE bookflow_db	0 row(s) affected
2	12:12:30	DROP TABLE customer_transactions	0 row(s) affected
3	12:13:24	SHOW TABLES	4 row(s) returned

DML Commands

2520030066

Mohammad Rehan

1. Insert Sample Data

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'bookflow_db' selected. The main editor window is in 'DDL' mode, showing the following SQL script:

```
3  txn_id INT PRIMARY KEY,
4  customer_name VARCHAR(50),
5  branch_name VARCHAR(50),
6  transaction_type VARCHAR(20),
7  amount DECIMAL(10,2),
8  transaction_date DATE
9  );
10
11  INSERT INTO bank_transactions VALUES
12  (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
13  (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
14  (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
15  (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
16  (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
17  (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
18  (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
19  (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
20  (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
21  (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
```

The 'Output' tab at the bottom shows the execution results:

#	Time	Action	Message
1	12:19:00	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1045. No database selected Select the default DE
2	12:20:03	USE bookflow_db	0 row(s) affected
3	12:20:05	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1050. Table 'bank_transactions' already exists
4	12:20:20	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sta','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

2. INSERT — New Transaction

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' tree with 'bookflow_db' selected. The main editor window is in 'DDL' mode, showing the following SQL script:

```
7  amount DECIMAL(10,2),
8  transaction_date DATE
9  );
10
11  INSERT INTO bank_transactions VALUES
12  (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
13  (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
14  (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
15  (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
16  (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
17  (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
18  (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
19  (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
20  (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
21  (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
22
23
24  INSERT INTO bank_transactions
25  VALUES
26  (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
```

The 'Output' tab at the bottom shows the execution results:

#	Time	Action	Message
1	12:19:00	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1045. No database selected Select the default
2	12:20:03	USE bookflow_db	0 row(s) affected
3	12:20:05	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1050. Table 'bank_transactions' already exists
4	12:20:20	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sta','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	12:24:58	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18')	1 row(s) affected

3. UPDATE

The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' tab is active, displaying 'No object selected'. The main editor contains the following SQL code:

```
26 (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
27
28 • UPDATE bank_transactions
29 SET amount = 5000
30 WHERE txn_id = 105;
```

The 'Output' tab shows the 'Action Output' with the following results:

#	Time	Action	Message
1	12:19:00	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1046. No database selected Select the default DB to t
2	12:20:03	USE bookflow_db	0 row(s) affected
3	12:20:05	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	Error Code: 1050. Table 'bank_transactions' already exists
4	12:20:20	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05), (102,'Sita','Hyd...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	12:24:58	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18)	1 row(s) affected
6	12:25:40	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

4. DELETE

The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' tab is active, displaying 'No object selected'. The main editor contains the following SQL code:

```
31
32 • DELETE FROM bank_transactions
33 WHERE txn_id = 111;
```

The 'Output' tab shows the 'Action Output' with the following results:

#	Time	Action	Message
2	12:20:03	USE bookflow_db	0 row(s) affected
3	12:20:05	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...	Error Code: 1050. Table 'bank_transactions' already exists
4	12:20:20	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05), (102,'Hyd...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	12:24:58	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18)	1 row(s) affected
6	12:25:40	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
7	12:26:46	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected

5. SELECT

The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' tab is active, displaying 'No object selected'. The main editor contains the following SQL code:

```
30 WHERE txn_id = 105;
31
32 • DELETE FROM bank_transactions
33 WHERE txn_id = 111;
34
35 • SELECT *
36 FROM bank_transactions;
```

The 'Result Grid' shows the following data:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
104	Anil	Vizag	Deposit	8000.00	2024-01-10
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16

The 'Output' tab shows the 'Action Output' with the following results:

#	Time	Action	Message
3	12:20:05	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...	Error Code: 1050. Table
4	12:20:20	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05), (102,'Sita','Hy...	10 row(s) affected Reco
5	12:24:58	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18)	1 row(s) affected
6	12:25:40	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows r
7	12:26:46	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected
8	12:27:35	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned

6. SELECT with WHERE

MySQL Workbench

Local instance MySQL80 x

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Navigator

SCHEMAS

Filter objects

bookflow_db

Tables

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

DML* x DDL

Limit to 1000 rows

```
34
35 SELECT *
36 FROM bank_transactions;
37
38 SELECT *
39 FROM bank_transactions
40 WHERE transaction_type = 'Deposit';
```

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Contents: I

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
104	Anil	Vizag	Deposit	8000.00	2024-01-10
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16

bank_transactions 2 x

Output

Action Output

#	Time	Action	Message
4	12:20:20	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyderabad','Deposit',7000,'2024-01-18')	10 row(s) affected Records: 11
5	12:24:58	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18')	1 row(s) affected
6	12:25:40	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows match
7	12:26:46	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected
8	12:27:35	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned
9	12:28:37	SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned

7. SELECT with ORDER BY

MySQL Workbench

Local instance MySQL80 x

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Navigator

SCHEMAS

Filter objects

bookflow_db

Tables

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

DML* x DDL

Limit to 1000 rows

```
38 SELECT *
39 FROM bank_transactions
40 WHERE transaction_type = 'Deposit';
41
42 SELECT *
43 FROM bank_transactions
44 ORDER BY amount DESC;
```

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Contents: I

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
104	Anil	Vizag	Deposit	8000.00	2024-01-10
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13

bank_transactions 3 x

Output

Action Output

#	Time	Action	Message
5	12:24:58	INSERT INTO bank_transactions VALUES (111,'Venu','Vizag','Deposit',7000,'2024-01-18')	1 row(s) affected
6	12:25:40	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows match
7	12:26:46	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected
8	12:27:35	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned
9	12:28:37	SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
10	12:29:30	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned

DCL & TCL COMMANDS

Mohammad Rehan

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DCL

1. GRANT SELECT and GRANT PRIVILEGES

The screenshot displays the MySQL Workbench interface with a SQL editor window titled 'DCL-TCL'. The editor contains the following SQL commands:

```
1 • USE bookflow_db;
2
3 • CREATE USER IF NOT EXISTS 'Auditor1'@'localhost'
4 IDENTIFIED BY 'Auditor@123';
5
6 • CREATE USER IF NOT EXISTS 'BranchManager'@'localhost'
7 IDENTIFIED BY 'Manager@123';
8
9 • GRANT SELECT
10 ON bookflow_db.bank_transactions
11 TO 'Auditor1'@'localhost';
12
13 • GRANT ALL PRIVILEGES
14 ON bookflow_db.bank_transactions
15 TO 'BranchManager'@'localhost';
16
17
```

The 'SCHEMAS' panel on the left shows the 'bookflow_db' database selected, with its tables, views, stored procedures, and functions listed. The 'Administration' and 'Schemas' tabs are visible at the bottom of the panel.

The 'Output' panel at the bottom shows the execution results of the commands:

#	Time	Action	Message
1	12:34:49	USE bookflow_db	0 row(s) affected
2	12:34:57	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'Auditor@123'	0 row(s) affected
3	12:35:00	CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'Manager@123'	0 row(s) affected
4	12:35:40	GRANT SELECT ON bookflow_db.bank_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
5	12:35:45	GRANT ALL PRIVILEGES ON bookflow_db.bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected

2. REVOKE SELECT AND REVOKE ALL PRIVELEGES

The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```
6 CREATE USER IF NOT EXISTS 'BranchManager'@'localhost'
7 IDENTIFIED BY 'Manager@123';
8
9 GRANT SELECT
10 ON bookflow_db.bank_transactions
11 TO 'Auditor1'@'localhost';
12
13 GRANT ALL PRIVILEGES
14 ON bookflow_db.bank_transactions
15 TO 'BranchManager'@'localhost';
16
17 REVOKE SELECT
18 ON bookflow_db.bank_transactions
19 FROM 'Auditor1'@'localhost';
20
21 REVOKE ALL PRIVILEGES
22 ON bookflow_db.bank_transactions
23 FROM 'BranchManager'@'localhost';
24
25
```

The Output window shows the execution results:

#	Time	Action	Message
2	12:34:57	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'Auditor@123'	0 row(s) affected
3	12:35:00	CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'Manager@123'	0 row(s) affected
4	12:35:40	GRANT SELECT ON bookflow_db.bank_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
5	12:35:45	GRANT ALL PRIVILEGES ON bookflow_db.bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
6	12:37:55	REVOKE SELECT ON bookflow_db.bank_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
7	12:38:15	REVOKE ALL PRIVILEGES ON bookflow_db.bank_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected

TCL

1. COMMIT & SAVEPOINT

The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```
12
13 GRANT ALL PRIVILEGES
14 ON bookflow_db.bank_transactions
15 TO 'BranchManager'@'localhost';
16
17 REVOKE SELECT
18 ON bookflow_db.bank_transactions
19 FROM 'Auditor1'@'localhost';
20
21 REVOKE ALL PRIVILEGES
22 ON bookflow_db.bank_transactions
23 FROM 'BranchManager'@'localhost';
24
25 UPDATE bank_transactions
26 SET amount = 6000
27 WHERE txn_id = 101;
28
29 COMMIT;
30
31 SAVEPOINT Before_Update;
```

The Output window shows the execution results:

#	Time	Action	Message
5	12:35:45	GRANT ALL PRIVILEGES ON bookflow_db.bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
6	12:37:55	REVOKE SELECT ON bookflow_db.bank_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
7	12:38:15	REVOKE ALL PRIVILEGES ON bookflow_db.bank_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
8	12:41:37	UPDATE bank_transactions SET amount = 6000 WHERE txn_id = 101	1 row(s) affected Rows n
9	12:41:39	COMMIT	0 row(s) affected
10	12:41:44	SAVEPOINT Before_Update	0 row(s) affected

2. ROLLBACK TO SAVEPOINT

MySQL Workbench

Local instance MySQL80 x

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Navigator

SCHEMAS

Filter objects

bookflow_db

Tables

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

No object selected

Object Info Session

DML DDL DCL-TCL*

Limit to 1000 rows

```
48
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57 • START TRANSACTION;
58
59 • SAVEPOINT Before_Update;
60
61 • UPDATE bank_transactions
62   SET amount = 99999
63   WHERE txn_id = 102;
64
65 • ROLLBACK TO SAVEPOINT Before_Update;
66
67 • COMMIT;
```

Output

Action Output

#	Time	Action	Message
14	12:43:24	ROLLBACK TO Before_Update	Error Code: 1305. SAVEPOINT Before_Update does not exist
15	12:44:10	START TRANSACTION	0 row(s) affected
16	12:44:12	SAVEPOINT Before_Update	0 row(s) affected
17	12:44:14	UPDATE bank_transactions SET amount = 99999 WHERE txn_id = 102	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0
18	12:44:16	ROLLBACK TO SAVEPOINT Before_Update	0 row(s) affected
19	12:44:18	COMMIT	0 row(s) affected